

# PRANAV JHA

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## Education

### Manipal University Jaipur

2022 - 2026

BTech(Hons) Computer Science and Engineering(IOT and Intelligent Systems)CGPA-7.61(3.27/4)

Jaipur, RJ

## Experience

- **InterraIT** Noida, Jan 2026 – Apr 2026  
*Student Trainee*
  - \* Developed a digital twin of the **FANUC R-50iA Robot Controller** using **NVIDIA Isaac Sim**.
  - \* Integrated **ROS2** for simulation-based testing, controller validation, and robotic performance analysis.
- **CSIR – National Institute of Oceanography** Goa, June 2025 – Aug 2025  
Research Intern
  - \* Used SegFormer, UNet, and PIDNet with Knowledge Distillation (Teacher–Student) to develop 19 lightweight real-time coral reef segmentation models.
  - \* Deployed 19 models on NVIDIA Jetson-Nano for live ROV-based underwater inference.
- **Siemens Digital Industries Software** Gurgaon, May 2024 – July 2024  
IoT Intern
  - \* Implemented real-time analytics on edge IoT devices using Azure AI.
  - \* Automated Excel workflows, reducing reporting time by 15%.

## Projects

- **Digital Twin of FANUC R-50i Robot Controller** Jan 2026 – Apr 2026  
*NVIDIA Isaac Sim, ROS2, FANUC*
  - \* Developed a digital twin of the FANUC R-50i Robot Controller using NVIDIA Isaac Sim.
  - \* Integrated ROS2 for simulation-based testing, validation, and performance analysis.
- **Coral Reef Semantic Segmentation System** June 2025 – Sep 2025  
*SegFormer, UNet, PIDNet, Jetson Nano*
  - \* Designed a lightweight semantic segmentation system using teacher–student distillation.
  - \* Deployed models on Jetson Nano for real-time underwater coral classification on ROV footage.
- **Real-Time Sign Language Detection System** Aug 2024 – Oct 2024  
*MediaPipe, TensorFlow, OpenCV*
  - \* Built a CNN-based model to recognize static and dynamic gestures using MediaPipe landmarks.
  - \* Enabled real-time sign-to-text translation for hearing- and speech-impaired users.

## Technical Skills

**Languages:** Python, Java, C, HTML, CSS, JavaScript, PHP\*, SQL

**Developer Tools:** VS Code, Eclipse, Arduino IDE, PyCharm, Mendix

**Technologies/Frameworks:** Linux, GitHub, Computer Vision, Machine Learning, Sci-kit Learn, Numpy, Pandas, PyTorch, TensorFlow, OpenCV, CUDA, Model Optimization, Knowledge Distillation IoT, Arduino, Raspberry Pi, Jetson-Nano, Deep Learning, Node-RED\*, ROS, Generative AI, 8051 *\*Elementary proficiency*

## Publications and Achievements

- **Research Publications:**
  - \* *Real-Time Coral Segmentation for Underwater ROVs* — Under Review (Journal & Conference), 2<sup>nd</sup> Author
  - \* *Autonomous Firefighting Robot in ROS-Gazebo* — Published, HITA 2025, 1<sup>st</sup> Author
  - \* *Hybrid Ensemble Models for Sustainability Prediction* — Accepted, COACONS 2025, 2<sup>nd</sup> Author
  - \* *Blockchain Optimization for Secure IoT (AMQP)* — Published, IEEE Xplore (ICAIHC 2025), 3<sup>rd</sup> Author
- **Patents:** Published 6 patents in Computer Vision, IoT, and Robotics.
- **Awards:** Received the **Excellence Award** 3 times for Outstanding Performance in Research and Technical Innovation.
- **Leadership:** Technical Secretary (3<sup>rd</sup> Year) and Head of Projects (2<sup>nd</sup> Year), IEEE Sensors Council, MUJ.
- **Outreach:** Invited back as an alumnus to deliver a robotics workshop, introducing school students to modern robotics through interactive lectures and hands-on projects.